

Ruifang (Terry) Zhang

☎ (+31) 626138070 ✉ zhangruifang0913@outlook.com 🌐 Ruifang (Terry) Zhang



Profile

Robotics M.Sc. student at TU Delft with experience in building and evaluating **robotic and autonomous systems**, including **perception, planning, and control pipelines**. Background in **system-level integration** from ADAS industry experience, working with **real-world data, testing workflows**, and **performance validation**. Interested in applying **robotics and automation** to practical engineering challenges in **industrial environments**.

Education

Delft University of Technology

M.Sc. IN ROBOTICS

Delft, Netherlands

Sep 2025 - Present

University College London (UCL)

M.Sc. IN ENGINEERING WITH ENTREPRENEURSHIP & INNOVATION

London, UK

Sep 2017 - Sep 2018

Yuan-Ze University

B.Sc. IN MECHANICAL ENGINEERING (GPA: 3.58/4.0, TOP 12%)

Taoyuan, Taiwan

Sep 2013 - Jun 2017

Project Experience

Radar-Camera 3D Object Detection with LiDAR-guided Distillation

TU Delft

Mar 2026

- Built a radar-dominant **3D object detection** pipeline on View-of-Delft with monocular image assisted point-level fusion and **CenterPoint-style BEV detection**.
- Implemented and debugged **cross-modal projection, feature sampling, fusion modules, LiDAR teacher training**, and **training-time distillation** under constrained compute budget.
- Designed reproducible ablations across baseline, **fusion, teacher**, and **distillation** settings, and improved best fusion performance.
- Demonstrated ability in **multimodal perception, PyTorch** model development, experimental design, debugging, and result analysis.

Carry-No-Spill | Shared-Control HRI Simulation Platform

TU Delft

Mar 2026

- Built a physics-based **HRI simulation platform** using **Python, pygame, Box2D (LiquidFun)**, and **esper ECS**, integrating **rigid-body dynamics, particle-based simulation**, and collision handling.
- Implemented modular control and experiment infrastructure, including **teleoperation, blind-mode evaluation**, runtime scoring, and automated logging of trajectories and task metrics.
- Developed procedural map generation and reproducible experiment workflows with fixed seeds, enabling quantitative comparison of **force-feedback** effects on performance.

Hierarchical Planning for Mobile Manipulation

TU Delft

Jan 2026

- **Pipeline Development:** Engineered a full **motion planning pipeline** for a non-holonomic robot in cluttered environments.
- **Hybrid Control & Planning:** Implemented **Informed RRT*** with kinodynamic steering and benchmarked **MPC vs. MPPI** for real-time control.
- **Quantitative Validation:** Validated system performance through **PyBullet** simulations, achieving improved time-to-goal while maintaining real-time feasibility.

Work Experience

Porsche China R&D Satellite

ADAS ADVISOR

Shanghai, China

Mar 2023 - Aug 2025

- Worked on **ADAS systems** involving **sensing, perception, and control pipelines** in real-world driving environments

2026-04-15

RUIFANG (TERRY) ZHANG · RESUME

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- Analyzed and benchmarked **sensor fusion** and **positioning systems**, identifying system-level performance gaps through vehicle testing
- Designed evaluation workflows including **scenario definition**, **data collection**, and **performance metrics** for system validation
- Collaborated with global engineering teams to align system design with region-specific requirements

Ford Research Engineering Center

Nanjing, China

ADAS PROJECT MANAGER / ENGINEER

Apr 2019 - Dec 2022

- Managed full lifecycle of **embedded systems** combining **sensing, perception, and control modules**
- Coordinated **system integration** across software, hardware, and validation teams (**HIL**, vehicle-level testing)
- Led debugging of system-level issues using **CAN tools** and diagnostic workflows
- Ensured **functional safety (FuSa)** and **cybersecurity compliance** for production systems

Technical Skills

Programming	C++, Python, MATLAB
Robotics & AI	ROS2, PyTorch, YOLO, Pybullet
Tools	Linux (Ubuntu), Git, Gazebo, Rviz, SolidWorks
Languages	English (Professional), Mandarin (Native), Dutch (Beginner)

Hobbies

Hiking (long-distance trails)
Tennis (2.0)
Piano (band keyboard player)